Physics-ML Neural Net For Aerodynamic Modeling

DrivAer Dataset | 2026

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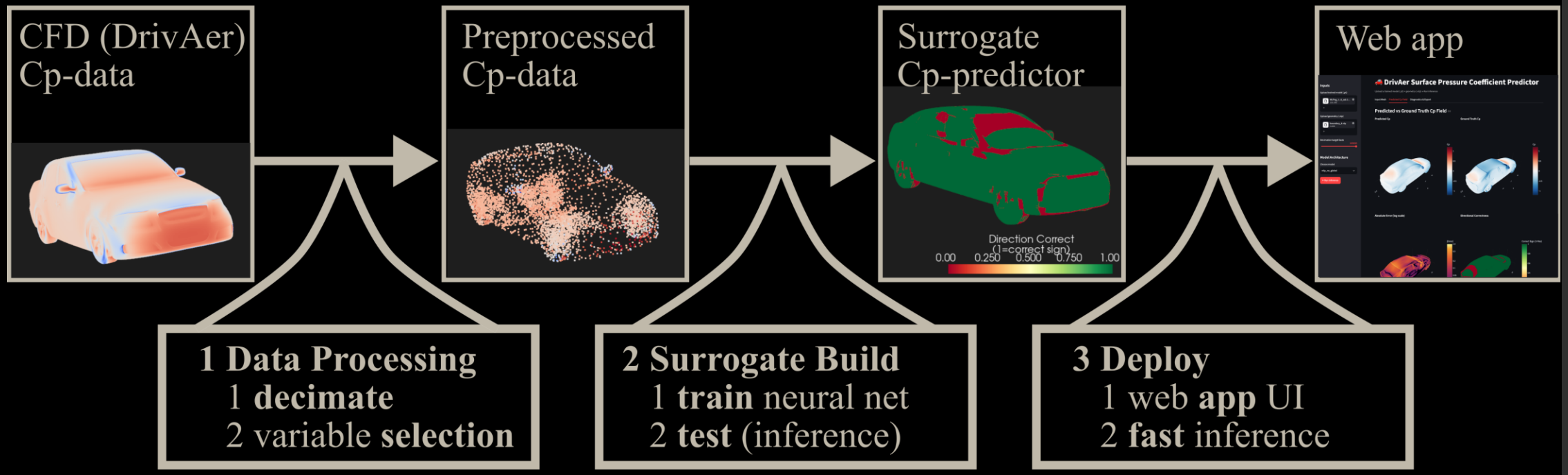


Traditional CFD is Too Slow In A Modern Design Cycle

- Aerodynamic design relies on Computational Fluid Dynamics (CFD)
- A single high-fidelity CFD simulation can take hours to days to complete
- Creates a bottleneck in the design process:
  - Limited number of design iterations
  - High computational cost
  - Delayed decision-making

Result : engineers must make critical design choices with incomplete information

CFD-Informed Neural Net Predicts Surface Pressure in Minutes (Prototype)



Prototype : point-based neural net (multilayer perceptron) trained on 10 surface geometries
Directionally correct (~90%) predictions in just a few minutes
Easy-to-use web app for rapid geometry evaluation

Fast, Actionable Insights That Accelerate Design

Speed: surface pressure **predictions in seconds** vs. hours/days with traditional CFD on new geometry

Usability: interactive **web app** allows engineers to upload geometries and instantly see predicted pressure fields

Performance: prototype shows strong directional accuracy (90%) on unseen geometries:

- Stagnation, Acceleration, Wake effects

Scalability: trained on only 10 cases

- Directionally correct (90% of the time).
- More training will increase accuracy and resolution.

Business Value:

- Enable 10x–100x more design iterations in early phases
- Reduce reliance on expensive CFD licenses for early exploration
- Support faster, more confident design decisions

Next Steps:

- Scale to larger datasets
- Uncertainty quantification